

χ -Theory: Control-Dependent Spectral Structure of Physical Systems

1 Introduction

The χ -framework proposes that physical systems are fundamentally defined not by state variables alone, but by relational structures of operational distinguishability.

In this view, observable physics emerges from spectral properties of an operator constructed from a distinguishability structure rather than directly from Hamiltonian dynamics.

The central claim is that different physical implementations of identical dynamics may lead to different observable behavior due to differences in underlying structure.

2 Fundamental Structure

A physical system is defined by the triple:

$$(\mathcal{D}, R, W)$$

where:

- $\mathcal{D}$ is a set of admissible states,
- R is a relation structure encoding allowed interactions,
- W_{ij} is a weight function encoding operational distinguishability between states i and j .

From this structure we define the distinguishability operator:

$$\hat{\Delta}_\chi = \mathcal{F}(\mathcal{D}, R, W),$$

where $\mathcal{F}$ is a structure-to-operator map.

3 Central Spectral Postulate

$$\boxed{\mathcal{O}_{phys} \sim \text{Spec}(\hat{\Delta}_\chi)}$$

Physical observables correspond to spectral properties of the distinguishability operator.

4 Control-Dependent Structure

Consider two implementations of the same Hamiltonian evolution:

$$H^{(1)} = H^{(2)}.$$

However, different experimental control sequences induce different distinguishability structures:

$$W^{(1)} \neq W^{(2)}.$$

This leads to:

$$\hat{\Delta}_\chi^{(k)} = \mathcal{F}(\mathcal{D}, R, W^{(k)}), \quad k = 1, 2.$$

5 Theorem

Control-Dependent Spectral Non-Equivalence

If two physical implementations share identical Hamiltonian dynamics but induce different distinguishability structures, then in general:

$$W^{(1)} \neq W^{(2)} \implies \text{Spec}(\hat{\Delta}_\chi^{(1)}) \neq \text{Spec}(\hat{\Delta}_\chi^{(2)})$$

even when:

$$H^{(1)} = H^{(2)}.$$

6 Proof (Structural Argument)

The result follows from three structural observations:

(1) Non-uniqueness of implementation

A given Hamiltonian evolution:

$$U(t) = \mathcal{T} \exp \left(-i \int H dt \right)$$

can be realized through multiple physically distinct control implementations.

(2) Control-induced structure

Each implementation induces a distinct effective distinguishability structure:

$$U^{(k)} \Rightarrow W^{(k)}.$$

Thus, identical dynamics does not imply identical relational structure.

(3) Spectral dependence

The distinguishability operator depends functionally on the induced structure:

$$\hat{\Delta}_\chi = \mathcal{F}(W), \quad \text{Spec}(\hat{\Delta}_\chi) = f(W).$$

Therefore structural inequivalence implies spectral inequivalence in general:

$$W^{(1)} \neq W^{(2)} \Rightarrow \text{Spec}(\hat{\Delta}_\chi^{(1)}) \neq \text{Spec}(\hat{\Delta}_\chi^{(2)}).$$

7 Variables

- H — Hamiltonian of the system
- W — operational distinguishability structure induced by control
- $\hat{\Delta}_\chi$ — distinguishability operator
- $\text{Spec}(\cdot)$ — spectrum of an operator
- $\mathcal{D}, R$ — state space and relational structure

8 Physical Consequence

Observable quantities such as coherence time, interference visibility, and decoherence rates depend not only on the Hamiltonian, but also on the control-induced structure W .

9 Experimental Regime

The theory becomes experimentally testable in systems where W is operationally controllable, such as trapped-ion quantum systems with different control implementations but identical Hamiltonian parameters.

In this regime:

$$W \rightarrow \hat{\Delta}_\chi \rightarrow \text{Spec}(\hat{\Delta}_\chi)$$

becomes experimentally accessible via coherence measurements.

10 Minimal Falsification Criterion

The framework is falsified if:

$$W^{(1)} \neq W^{(2)} \quad \text{but} \quad \text{Spec}(\hat{\Delta}_\chi^{(1)}) = \text{Spec}(\hat{\Delta}_\chi^{(2)})$$

within experimental precision.

11 Conclusion

The χ -framework establishes a structural link between control, distinguishability, and spectral physics, suggesting that identical Hamiltonian dynamics may yield distinct observable physics due to differences in underlying control-induced structure.